CBSE Class 10 Mathematics Standard Question Paper 2026 Set 30/4/2

Time Allowed :3 Hours	**Maximum Marks :**80	**Total Questions :**38

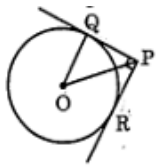
General Instructions

Read the following instructions very carefully and strictly follow them:

1. *This question paper hass **38** questions. **All** questions are compulsory.*
2. *This question paper contains **five** sections - Section **A, B, C, D** and **E**.*
3. ***Section A** - Questions no. **1** to **18** are Multiple Choice Type Questions (MCQs) and questions number 19 and 20 are Assertion-Reason based questions of 1 mark each.*
4. ***Section B** - Questions no. **21** to **25** are Very-Short Answer (VSA) Type Questions, carrying 2 marks each.*
5. ***Section C** - Questions no. **26** to **31** are Short-Answer (SA) Type Questions, carrying 3 marks each.*
6. ***Section D** - Questions no. **32** to **35** are Long-Answer (LA) Type Question, carrying 5 marks each.*
7. ***Section E** - Questions no. **36** to **38** are case study based questions carrying 4 marks each. Internal choice is provided in 2 marks questions in each case study.*
8. *There is no overall choice given in the question paper. However, an internal choice has been provided in 2 questions in Sections B, 2 questions in Section C, 2 questions in Section D and 3 questions in Section E.*
9. *Use of calculator is **not** allowed.*
10. *Please write down the serial number of the question in the answer-book at the given place before attempting it.*
11. *15 minute time has been allotted to read this question paper. The question paper will be distributed at 10:15 a.m. From 10:15 a.m. to 10:30 a.m., the candidates will read the question paper only and will not write any answer on the answer-book during this period.*

Section - A

1. If PQ and PR are tangents to the circle with centre O and radius 4 cm such that $\angle QPR = 90^\circ$, then the length OP is



(A) 4 cm
(B) $4\sqrt{2}$ cm
(C) 8 cm
(D) $2\sqrt{2}$ cm

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2. An ice-cream cone of radius r and height h is completely filled by two spherical scoops of ice-cream. If radius of each spherical scoop is $\frac{r}{2}$, then $h : 2r$ equals
(A) $1 : 8$
(B) $1 : 2$
(C) $1 : 1$
(D) $2 : 1$

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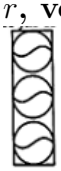
3. Arc PQ subtends an angle θ at the centre of the circle with radius 6.3 cm. If Arc $PQ = 11$ cm, then the value of θ is
(A) 10°
(B) 60°
(C) 45°
(D) 100°

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4. $\frac{1+\tan^2 A}{1+\cot^2 A}$ equals to:
(A) $\tan^2 A$
(B) -1
(C) $-\tan^2 A$
(D) $\cot^2 A$

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5. Three tennis balls are just packed in a cylindrical jar. If radius of each ball is r, volume of air inside the jar is



(A) $2\pi r^3$
(B) $3\pi r^3$
(C) $5\pi r^3$
(D) $4\pi r^3$

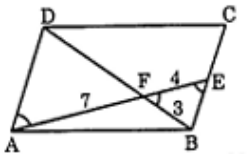
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6. Two different dice are rolled together. The probability that both the obtained numbers are less than 4, is

(A) $\frac{2}{9}$
(B) $\frac{7}{36}$
(C) $\frac{1}{4}$
(D) $\frac{2}{3}$

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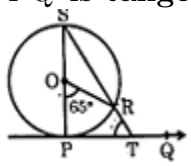
7. $ABCD$ **is a parallelogram such that** $AF = 7$ **cm,** $FB = 3$ **cm and** $EF = 4$ **cm, length** FD **equals**



(A) $\frac{21}{4}$ cm
(B) $\frac{28}{3}$ cm
(C) $\frac{12}{7}$ cm
(D) 5.5 cm

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8. PQ **is tangent to a circle with centre** O**. If** $\angle POR = 65°$**, then** $m\angle OPR$ **is**



(A) 65°
(B) 58.5°
(C) 57.5°
(D) 45°

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9. A circle centred at $(-1, 2)$ **passes through the point** $(0, 3)$**. Radius of the circle is**

(A) $2\sqrt{2}$
(B) $\sqrt{2}$
(C) $\sqrt{26}$
(D) 1

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10. It is given that $\triangle ABC \sim \triangle EDF$**. Which of the following is not true?**

(A) $\frac{\text{Perimeter of } \triangle ABC}{\text{Perimeter of } \triangle EDF} = \frac{AB}{ED}$
(B) $\frac{AB}{ED} = \frac{AC}{EF}$
(C) $\angle A = \angle D, \angle C = \angle F$
(D) $\frac{AB+BC}{AC} = \frac{ED+DF}{EF}$

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11. If roots of the quadratic equation $x^2 - k\sqrt{3}x + 2 = 0$ are real and equal, then value of k is

(A) -2

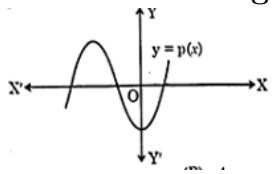
(B) $\sqrt{\frac{8}{3}}$

(C) 1

(D) 2

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12. Observe the graph of polynomial $p(x)$. Number of zeroes of $p(x)$ is



(A) 5

(B) 4

(C) 6

(D) 3

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13. Mean and Median of a frequency distribution are 43 and 40 respectively. The value of mode is

(A) 34

(B) 43

(C) 38.5

(D) 41.5

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14. Area of sector of a circle with radius 18 cm is 198 cm^2. The measure of central angle is

(A) $70°$

(B) $14°$

(C) $140°$

(D) $210°$

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15. If $2\tan A = 3$, then value of $\sec A$ equals

(A) $\frac{\sqrt{13}}{2}$

(B) $\frac{\sqrt{13}}{4}$

(C) $\frac{2}{\sqrt{13}}$

(D) $\frac{\sqrt{13}}{2}$

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16. The value of k for which the system of linear equations $\frac{x}{2} + \frac{y}{3} = 5$ and $2x + ky = 7$ is inconsistent, is

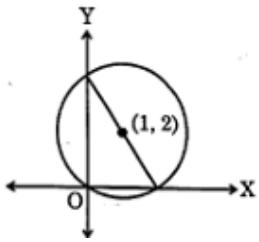
(A) $\frac{3}{4}$
(B) $\frac{4}{3}$
(C) $\frac{1}{3}$
(D) 3

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17. In an A.P., $a = -3$ and $S_{17} = 357$. The value of a_{17} is
(A) 47
(B) 39
(C) 45
(D) 42

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18. In the given figure, a circle is centred at $(1, 2)$. The diameter of the circle is



(A) 4
(B) $2\sqrt{2}$
(C) $\sqrt{5}$
(D) $2\sqrt{5}$

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19. Assertion (A) : $(\sqrt{3} + \sqrt{5})$ is an irrational number.
Reason (R) : Sum of the any two irrational numbers is always irrational.
(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

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20. Assertion (A) : If probability of happening of an event is $0.2p$, $p > 0$, then p can't be more than 5.
Reason (R) : $P(\bar{E}) = 1 - P(E)$ for an event E.
(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

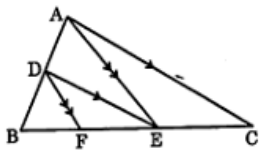
Section - B

21(a). Prove that $2 + 3\sqrt{5}$ is an irrational number given that $\sqrt{5}$ is an irrational number.

21(b). If the HCF of 210 and 55 is expressed as $210 \times 5 + 55m$, then find the value of m.

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22. In the given figure, $DE \parallel AC$ and $DF \parallel AE$. Prove that : $\frac{BF}{FE} = \frac{BE}{EC}$.



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23. Verify that roots of the quadratic equation $(p - q)x^2 + (q - r)x + (r - p) = 0$ are equal when $q + r = 2p$.

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24. α, β are zeroes of the polynomial $p(x) = 3x^2 - 6x - 5$. Find the value of $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$.

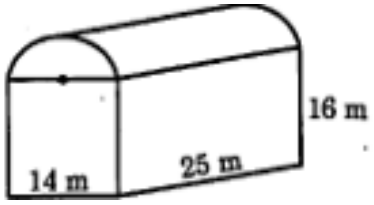
25(a). Prove that : $\sqrt{\frac{1+\sin A}{1-\sin A}} = \sec A + \tan A$.

25(b). Evaluate : $\frac{3\cos^2 30° - 6\csc^2 30°}{\tan^2 60°}$.

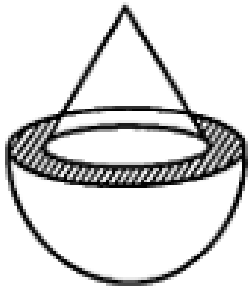
Section - C

26. A trader has three different types of oils of volume 870 l, 812 l and 638 l. Find the least number of containers of equal size required to store all the oil without getting mixed.

27(a). To protect plants from heat, a shed of iron rods covered with green cloth is made. The lower part of the shed is a cuboid mounted by semi-cylinder as shown in the figure. Find the area of the cloth required to make this shed, if dimensions of the cuboid are 14 m $\times$ 25 m $\times$ 16 m.



27(b). The internal and external radii of a hollow hemisphere are $5\sqrt{2}$ cm and 10 cm respectively. A cone of height $5\sqrt{7}$ cm and radius $5\sqrt{2}$ cm is surmounted on the hemisphere as shown in the figure. Find the total surface area of the object in terms of π. (Use $\sqrt{2} = 1.4$)



28(a). In a class test, Veer scored 6 more than twice as many marks as Kevin scored. If one of them had scored 4 more marks, their total score would have been 40. Find the marks obtained by Veer and Kevin.

28(b). Solve the linear equations $3x + y = 14$ and $y = 2$ graphically.

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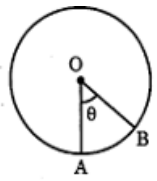
29. A bag contains 30 balls out of which 'm' number of balls are blue in colour.
(i) Find the probability that a ball drawn at random from the bag is not blue.
(ii) If 6 more blue balls are added in the bag, then the probability of drawing a blue ball will be $\frac{5}{4}$ times the probability of drawing a blue ball in the first case. Find the value of m.

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30. Prove that : $\frac{1}{\sec x - \tan x} - \frac{1}{\cos x} = \frac{1}{\cos x} - \frac{1}{\sec x + \tan x}$

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31. The perimeter of sector OAB of a circle with centre O and radius 5.6 cm, is 15.6 cm. Find length of the arc AB. Also find the value of θ.



Section - D

32. A kite is flying at a height of 60 m above the ground level. Ravi, standing at the roof of the house is holding the string straight and observes the angle of elevation of kite as $30°$. From the bottom of the same building, the angle of elevation of kite is $45°$. Find the length of the string and height of roof from the ground. (Use $\sqrt{3} = 1.73$)

33(a). Find mean and mode of the following frequency distribution :

Class :	5 - 15	15 - 25	25 - 35	35 - 45	45 - 55	55 - 65
Frequency :	11	20	25	22	12	10

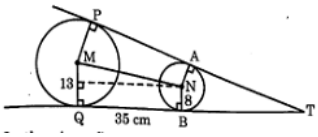
33(b). The median of the following data is 32.5, find the missing frequencies x and y :

Class :	0-10	10-20	20-30	30-40	40-50	50-60	60-70	Total
Frequency :	x	5	9	12	y	3	2	40

34(a). A person on tour has ₹5,400 for his expenses. If he extends his tour by 5 days, he has to cut down his daily expenses by ₹180. Find the original duration of the tour and daily expense.

34(b). The total cost of certain piece of cloth was ₹2,100. During special sale time, the shopkeeper offered 2 m extra cloth for free thus reducing the price of cloth per metre by ₹120. What was the original per metre price of cloth and its length?

35.

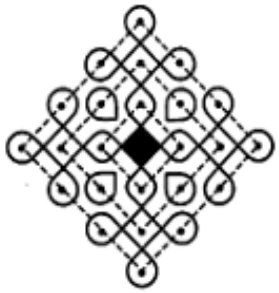


In the given figure, TP and TQ are tangents to a circle with centre M, touching another circle with centre N at A and B respectively. It is given that $MQ = 13$ cm, $NB = 8$ cm, $BQ = 35$ cm and $TP = 80$ cm.

(i) Name the quadrilateral MQBN. (1)
(ii) Is MN parallel to PA? Justify your answer. (1)
(iii) Find length TB. (1)
(iv) Find length MN. (2)

Section - E

36. 'Kolam' is a decorative art which is made with rice flour in South Indian States. It is drawn on grid pattern of dots. One such art work is shown below.



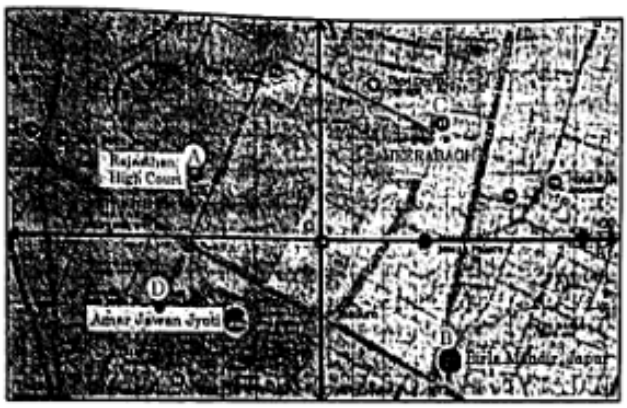
Observe the given figure carefully. There are 4 dots in first square, 8 dots in second square, 12 dots in third square and so on. Based on the above, answer the following questions:
36(i). Show that number of dots given above form an A.P. Write the first term and common difference.

36(ii). Write n^{th} term of the A.P. formed.

36(iii)(a). The pattern is expanded on a large ground. If total 220 dots are used, then find the number of squares formed.

36(iii)(b). Is it possible to complete n number of squares using 100 dots? If yes, then find the value of n.

37.



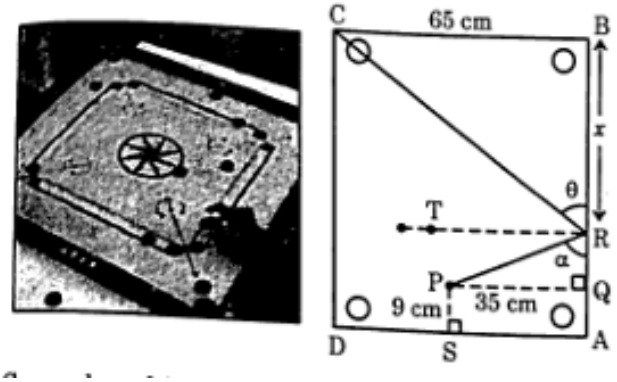
Observe the map of Jaipur city placed on a Cartesian plane. Taking Rambagh Palace as origin, the location of some places are given below:
Point A: $(-4, 2)$ **Rajasthan High Court**
Point B: $(4, -4)$ **Birla Mandir**
Point C: $(4, 3)$ **Heera Bagh**
Point D: $(-5, -2)$ **Amar Jawan Jyoti**
Based on the above, answer the following questions:
37(i). Advocate Rehana stays at Heera Bagh. How much distance she has to cover daily to go to the court and coming back home?

37(ii). There is a crossing on X-axis which divides AD in a certain ratio. Find the ratio.

37(iii)(a). Is Birla Mandir equidistant from Heera Bagh and Amar Jawan Jyoti? Justify your answer.

37(iii)(b). Using section formula, show that points A, O and B are not collinear.

38.



Carom board is a very popular game. The board is a square of side length 65 cm. It has circular pockets in each corner. Ansh strikes a disc, kept at position P with

a striker. The disc, hits the boundary of the board at R and goes straight to pocket at corner C. It is given that $PS = 9$ cm, $PQ = 35$ cm, $BR = x$, $\angle PRQ = \alpha$ and $\angle CRB = \theta$. Based on the above information, answer the following questions:

38(i). Using law of reflection i.e. $\angle PRT = \angle CRT$, prove that $\theta = \alpha$.

38(ii). Prove that $\triangle PQR \sim \triangle CBR$ given that PQ is perpendicular to AB.

38(iii)(a). Find the value of x using similarity of triangles.

38(iii)(b). If $\frac{\textbf{Area } \triangle PQR}{\textbf{Area } \triangle CBR} = \frac{PQ^2}{CB^2}$, then find the value of x.
